

# Statistics & Excel Data Analysis

## Part 1 → The Most Important Differences Beginner-Friendly Summary and Decision Guide

### How to use this PDF

Do not start by memorizing formulas. First ask: **What question am I trying to answer?** Then choose the method. This guide explains the meaning, use, and easiest memory trick for each method.

## PART 1 — BASIC STATISTICS

### 1. Descriptive Statistics

Meaning: Descriptive Statistics summarizes your data.

It gives information such as mean, median, mode, minimum, maximum, variance and standard deviation.

Use it when you ask: “What does my data look like?”

Easy memory: DESCRIPTIVE = Describe my data.

Example: Marks = 50, 60, 70, 80, 90. Mean =  $(50+60+70+80+90)/5 = 70$ .

### 2. Histogram

Meaning: A Histogram shows how numerical values are distributed.

It groups values into intervals/bins and shows how many observations fall in each interval.

Use it when you ask: “Where are most of my values?”

Easy memory: HISTOGRAM = Frequency distribution.

Example: 0–20 → 2 students, 20–40 → 8, 40–60 → 15, etc.

### 3. Rank and Percentile

Meaning: Rank gives the position of a value; percentile gives its relative position compared with other observations.

Rank example: highest mark = Rank 1. Percentile example: 90th percentile means the value is higher than approximately 90% of observations.

Use it for competition, student ranking, employee ranking, or comparing performance.

Easy memory: RANK = Position; PERCENTILE = Relative position.

### 4. Random Number Generation

Meaning: Creates random values.

It can generate random numbers within a chosen range or according to a selected distribution.

Use it for simulation, testing, probability experiments, and creating practice data.

Easy memory: RANDOM NUMBER = Create random data.

### 5. Sampling

Meaning: Sampling means selecting a smaller group from a larger population.

Population = the whole group. Sample = a smaller part selected from that group.

Use it when studying the entire population is too costly, slow, or difficult.

Easy memory: POPULATION = Big group; SAMPLE = Small part.

Example: 10,000 students are the population; 500 selected students are the sample.

## PART 2 — RELATIONSHIP BETWEEN VARIABLES

### 6. Covariance

Meaning: Covariance tells whether two variables tend to move together.

Positive covariance: X increases and Y tends to increase. Negative covariance: X increases and Y tends to decrease.

Near zero: no clear linear co-movement.

Use it when asking: "Do these two variables move together?"

Easy memory: COVARIANCE = Direction of movement.

Example: Study hours increase and marks tend to increase → positive covariance.

### 7. Correlation

Meaning: Correlation measures both the direction and strength of a linear relationship.

Its value is generally between -1 and +1. +1 = perfect positive, 0 = no linear relationship, -1 = perfect negative.

Use it when asking: "How strongly are these two variables related?"

Easy memory: CORRELATION = Direction + Strength.

Example: Correlation = 0.95 → very strong positive linear relationship.

### 8. Regression

Meaning: Regression is used to predict one variable using one or more other variables.

X = independent/input variable. Y = dependent/output variable. A regression equation can estimate Y from X.

Use it when asking: "Can I predict Y using X?"

Easy memory: REGRESSION = Prediction.

Example: Advertising spending → Sales. Use advertising spending to predict sales.

## PART 3 — FORECASTING / TIME SERIES

### 9. Moving Average

Meaning: A moving average calculates an average over a fixed number of recent observations and then moves the window forward.

It reduces short-term fluctuations and makes the underlying trend easier to see.

Use it for sales, demand, monthly data, daily data, and other time-series smoothing.

Easy memory: MOVING AVERAGE = Smooth the data.

Example: 3-value average =  $(100+120+140)/3 = 120$ .

### 10. Exponential Smoothing

Meaning: Exponential smoothing forecasts by giving more importance to recent observations.

Unlike a simple moving average, recent data gets greater weight.

Use it when forecasting and recent observations should influence the forecast more strongly.

Easy memory: EXPONENTIAL SMOOTHING = Recent data gets more importance.

## PART 4 — COMPARING VARIANCES

### 11. F-Test Two-Sample for Variances

Meaning: The F-test compares the variances/spread of two groups.

It helps determine whether the variability of two groups is significantly different.

Use it when asking: “Are the variances of two groups different?”

Easy memory: F-TEST = Compare variances.

Example: Compare the variation in production from Machine A and Machine B.

## **PART 5 — T-TESTS**

### **12. t-Test: Paired Two Sample for Means**

Meaning: Compares two measurements from the same subjects/items.

The observations are paired because the same person/item is measured twice.

Use it for before-vs-after studies, such as before training vs after training.

Easy memory: PAIRED = Same people/items twice.

Example: Student marks before training and after training.

### **13. t-Test: Two-Sample Assuming Equal Variances**

Meaning: Compares the means of two independent groups while assuming equal population variances.

The groups contain different/independent subjects.

Use it for two separate groups when the equal-variance assumption is appropriate.

Easy memory: 2 GROUPS + EQUAL VARIANCE = Equal-variance t-test.

Example: Class A vs Class B with equal variance assumption.

### **14. t-Test: Two-Sample Assuming Unequal Variances**

Meaning: Compares the means of two independent groups without assuming equal variances.

This is commonly known as Welch's two-sample t-test.

Use it when the two groups may have different variability.

Easy memory: 2 GROUPS + UNEQUAL VARIANCE = Unequal-variance t-test.

Example: Compare two groups where one group is much more spread out than the other.

## **PART 6 — Z-TEST**

### **15. z-Test: Two Sample for Means**

Meaning: Compares the means of two groups using the z distribution under appropriate z-test assumptions.

A common situation is when the relevant population variance is known; large-sample normal approximations can also justify z procedures.

Use it when the z-test assumptions are satisfied.

Easy memory: z-TEST = Compare means using z.

## **PART 7 — ANOVA**

### **16. ANOVA: Single Factor**

Meaning: ANOVA (Analysis of Variance) is used to compare the means of three or more groups using one factor.

Example factor: teaching method with Method A, B, and C.

Use it when asking: “Are the average values of 3+ groups different?”

Easy memory: SINGLE FACTOR = 1 factor, 3+ groups.

### **17. ANOVA: Two-Factor With Replication**

Meaning: Uses two factors and multiple observations for each combination.

It can examine the effect of Factor 1, Factor 2, and their interaction.

Use it when each factor combination has repeated observations.

Easy memory: 2 FACTORS + REPEATED OBSERVATIONS = With Replication.

### **18. ANOVA: Two-Factor Without Replication**

Meaning: Uses two factors but only one observation for each combination.

There are no repeated observations for each factor combination.

Use it when the data table has one value per combination of the two factors.

Easy memory: 2 FACTORS + ONE OBSERVATION PER CELL = Without Replication.

## **PART 8 — FOURIER ANALYSIS**

### **19. Fourier Analysis**

Meaning: Fourier Analysis breaks a complicated signal/time series into frequency-based components.

It is useful for detecting periodic or repeating patterns.

Use it in signal processing, audio, engineering, and advanced time-series analysis.

Easy memory: FOURIER = Find repeating/cyclic patterns.

For a beginner, learn descriptive statistics, correlation, regression, and basic time series before Fourier.

# THE MOST IMPORTANT DIFFERENCES

This is the final quick-reference section. Learn these differences first.

| Method                                  | Main Question                               | When to Use                                                                                 | Easy Memory             |
|-----------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------|-------------------------|
| Descriptive Statistics                  | What does my data look like?                | Summarize mean, median, SD, min, max, etc.                                                  | Describe                |
| Histogram                               | How are values distributed?                 | See frequency/distribution of numerical data.                                               | Distribution            |
| Rank vs Percentile                      | What is the position?                       | Rank = position; Percentile = relative position.                                            | Position                |
| Covariance vs Correlation               | Do two variables move together?             | Covariance = direction; Correlation = direction + strength (-1 to +1).                      | Relate                  |
| Correlation vs Regression               | Relationship or prediction?                 | Correlation measures relationship; Regression predicts Y from X.                            | Relate vs Predict       |
| Moving Average vs Exponential Smoothing | How do I smooth/forecast time data?         | Moving average uses a fixed window; exponential smoothing gives more weight to recent data. | Smooth vs Recent weight |
| F-Test                                  | Are two variances different?                | Compare spread/variance of two groups.                                                      | Variance                |
| Paired t-Test                           | Did the same subjects change?               | Same people/items measured twice.                                                           | Same group twice        |
| 2-Sample t-Test — Equal Variances       | Are two independent means different?        | Two separate groups; equal variance assumption.                                             | 2 groups + equal        |
| 2-Sample t-Test — Unequal Variances     | Are two independent means different?        | Two separate groups; unequal variance allowed.                                              | 2 groups + unequal      |
| z-Test — Two Sample Means               | Are two means different?                    | Use when z-test assumptions apply.                                                          | Means + z               |
| ANOVA Single Factor                     | Are 3+ group means different?               | One factor with 3+ groups.                                                                  | 3+ groups, 1 factor     |
| ANOVA 2-Factor With Replication         | Do two factors affect the result?           | Two factors + repeated observations.                                                        | 2 factors + repeats     |
| ANOVA 2-Factor Without Replication      | Do two factors affect the result?           | Two factors + one observation per combination.                                              | 2 factors + no repeats  |
| Sampling                                | Can I study a smaller part of a population? | Use a sample instead of the full population.                                                | Big → Small             |
| Random Number Generation                | Can I create random data?                   | Simulation and testing.                                                                     | Random data             |
| Fourier Analysis                        | Are there repeating/cyclic patterns?        | Advanced signal/time-series analysis.                                                       | Find cycles             |

## BEST DECISION METHOD

1. Want to **describe** your data? → Descriptive Statistics
2. Want to see **distribution**? → Histogram
3. Want **position**? → Rank / Percentile
4. Want to know whether two variables **move together**? → Covariance
5. Want the **strength of relationship**? → Correlation
6. Want to **predict** Y from X? → Regression

7. Want to **smooth** time-series data? → Moving Average
8. Want a forecast where **recent data matters more**? → Exponential Smoothing
9. Want to compare **two variances**? → F-Test
10. Want to compare **two means**? → t-Test / z-Test
11. Same people/items measured twice? → **Paired t-Test**
12. Two independent groups? → **Two-Sample t-Test**
13. Two independent groups + equal variance assumption? → **Equal-Variance t-Test**
14. Two independent groups + unequal variances? → **Unequal-Variance t-Test**
15. Three or more group means? → **ANOVA**
16. Two ANOVA factors + repeated observations? → **With Replication**
17. Two ANOVA factors + one observation per combination? → **Without Replication**
18. Looking for repeating cycles? → **Fourier Analysis**

**7-WORD MEMORY:** DESCRIBE → DISTRIBUTE → RELATE → PREDICT → FORECAST → COMPARE → TEST

**Most important:** Correlation = relationship; Regression = prediction; F-Test = variance; t/z-Test = means; ANOVA = 3+ means.